RTU — UNIVERSAL THERMODYNAMIC REVOLUTION

An Ultra Low-Cost Solar Energy Generation System Built from Urban Waste: Scalable Five-Layer Architecture for Thermal, Electric and Refrigeration Output

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"The sun shines for everyone"

Fig. 1 — RTU prototype in field conditions, Córdoba. PET-aluminium collector with integrated reservoir.

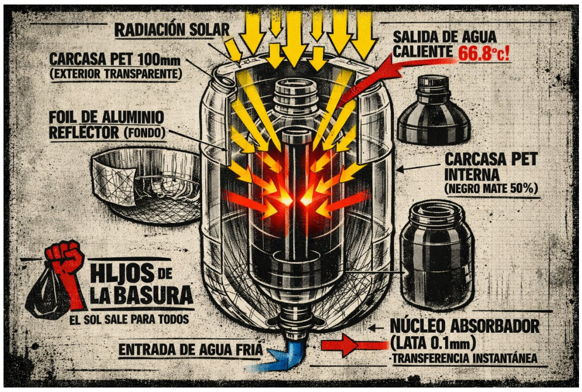

Fig. 2 — RTU tube cross-section diagram: PET layers · 0.1mm can · Tetra Brik rear reflector.

Keywords: *ultra low-cost solar energy, waste-to-energy, urban mining, recycled solar collector, PET bottle collector, aluminium can absorber, Tetra Brik reflector, low-tech ORC cycle, adaptive dynamic expansion, Arduino thermal control, 24/7 continuous electricity, expansion cooling, data centre refrigeration, energy sovereignty, open-source hardware, circular economy, five-layer platform, social thermodynamics, tropical countries, global south, scalable off-grid*

ABSTRACT

We present the RTU (Universal Thermodynamic Revolution) system: an ultra low-cost solar energy platform built entirely from urban waste, capable of simultaneously delivering domestic hot water, thermal energy storage, continuous 24/7 electricity, passive refrigeration, and local

employment — powered exclusively by solar radiation and constructed without a single component that is not recovered urban waste.

The collector is built from transparent PET bottles as heat traps, 0.1mm aluminium cans as instantaneous-transfer absorber cores, and the metallised interior of Tetra Brik cartons as integrated rear reflectors — a key innovation that eliminates the last purchased material from the design, achieving a 100% recycled collector. Two independent field measurements in Córdoba, Argentina (lat. 31°S) documented 177 W/m² net thermal power and a maximum output temperature of 80°C in April — exceeding commercial flat-plate collectors and reaching the operational range of evacuated tube systems.

The thermal reservoir repurposes the insulated cabinet of a discarded household refrigerator. In urban practice, the compressor and refrigerant circuit are removed by the informal scrap metal market within hours of street disposal, leaving a carcass with intact polyurethane insulation ($\lambda \approx$ 0.022–0.028 W/(m·K)) — industrial-grade thermal storage at zero cost. The system manages circulation via Arduino Mega with a DS18B20 sensor network and a photovoltaic cell as solar radiation sensor, implementing adaptive differential control with hysteresis and PWM proportional to instantaneous irradiation.

A theoretically grounded but experimentally unvalidated Layer 3 proposes continuous electrical generation via a low-tech Organic Rankine Cycle with a critical innovation: an adaptive dynamic expansion system governed by solenoid valves and Arduino Nano, which continuously adjusts the active volume of the thermodynamic cycle according to the available thermal differential in real time. This mechanism is what enables the design estimate of 30 continuous electrical watts per square metre of collector over 24 hours, regardless of temperature fluctuations throughout the day-night cycle. Validation of this layer requires refrigeration engineering expertise and represents the primary objective of the next funding phase.

A waste-stream analysis for Argentina indicates that annual production of PET bottles (3.76 billion), aluminium cans (1.41 billion), and Tetra Brik cartons (2.44 billion) could support up to 39.2 million m² of collector area. At 30 W/m² continuous electrical output, this represents 1.2 GW of continuous power — 10.3 TWh/year — equivalent to 8% of Argentina's total electricity consumption, generated entirely from currently discarded urban waste. In tropical regions (Sahel, India, NE Brazil), irradiation factors multiply available output by up to ×1.25, with full system payback of 3–4 months at local electricity tariffs.

The fifth and final layer of the RTU platform is social: the barrier to energy transition in low-income communities is not technological or material — both are resolved by this system. It is organisational. Institutions with territorial presence, training capacity and community service orientation are the natural replication vector. Three days of training, waste materials and this document are sufficient to deploy energy sovereignty in any sun-accessible community worldwide.

1. INTRODUCTION: THE PARADOX OF ABUNDANCE

The contemporary energy crisis is not solely a crisis of electricity generation. It is simultaneously a crisis of access, of storage, of material distribution and, fundamentally, of social organisation. While billions of people lack adequate energy services, cities worldwide accumulate vast quantities of discarded industrial materials that retain most of their original functional value.

Access to domestic hot water represents 25–40% of residential energy expenditure in Argentina and Latin America. Commercial solar thermal systems (USD 400–4,400) and photovoltaic systems with battery storage for 24-hour operation (USD 800–3,000) remain systematically beyond the reach of lower-income sectors. The RTU system resolves both problems simultaneously using urban waste and low-cost electronics.

This work is organised around a central question: **Can a city become its own mine?**

2. URBAN ENERGY MINING

Conventional mining extracts resources from nature, degrading the environment. Urban mining extracts resources from civilisation itself, rehabilitating it. Every discarded refrigerator contains industrial thermal insulation, heat exchangers and metallic structures. Every PET bottle contains already-refined raw material. Every aluminium can embodies previously invested metallurgical energy. From this perspective, waste ceases to be the tragic end of an economic cycle and becomes the beginning of a new energy cycle.

2.1 The real lifecycle of a discarded refrigerator

Contrary to formal regulation, in Argentine urban practice a refrigerator placed on the street is stripped of its compressor — which has copper and steel scrap value — by informal waste collectors within hours. The remaining carcass is already free of refrigerant gases and compressor oil. There is no legal or technical barrier to its direct reuse. The injected polyurethane foam insulation remains fully intact: $\lambda \approx 0.022$–0.028 W/(m·K), industrial grade, zero cost.

Component	Status in abandoned carcass	Role in RTU
Refrigerant gas	Absent — removed with compressor by scrap collector	No legal restriction
Compressor + oil	Absent — scrap metal value extracted within hours	No legal restriction
Polyurethane foam	**Present and INTACT**	**KEY ASSET: thermal insulation Layer 2 reservoir**
Serpentine / Al·Cu plate	**Present**	**KEY ASSET: ORC evaporator Layer 3 cycle**
Metal carcass	Present	Reservoir structure

Table 1. Status of discarded refrigerator components and their role in the RTU system. The informal scrap market effectively pre-processes the WEEE, removing the only components that would require licensed handling.

3. RTU ARCHITECTURE — FIVE FUNCTIONAL LAYERS

The RTU platform is organised in five functional layers of escalating complexity. Layers 1 and 2 are experimentally validated. Layers 3 and 4 are theoretically grounded design concepts, pending experimental validation. Layer 5 is the inevitable social consequence of the preceding four.

Layer	Function	Output	Status	Complexity
1	**Direct heat production**	**Hot water up to 80°C**	**Experimentally validated**	★■■

2	**Thermal battery (insulated reservoir)**	**9.3 kWh/200L 2–4°C loss/night**	**Experimentally validated**	★■■
3	**Electrical generation ORC adaptive expansion**	**30W continuous per m² · 24/7**	**Theoretical design requires investment**	★★★
4	**Residual cold from expansion**	**Reusable refrigeration**	**Theoretical design requires investment**	★★★
5	Social Thermodynamics employment + autonomy	Distributed local industry	Conceptual framework	—

Table 2. RTU five-layer architecture. Green: experimentally validated. Yellow: theoretical design.

4. LAYER 1 — PET-ALUMINIUM-TETRA BRIK SOLAR COLLECTOR

Fig. 3 — Complete RTU system: PET-aluminium tube collector, insulated upper header, active thermosiphon and integrated reservoir. Documented output: 80°C (April).

4.1 Multi-layer tube architecture

Layer	Material	Physical Function
1 — Outer shell	Transparent PET bottle 100mm diameter	Convective barrier; local greenhouse effect; maximum transmission of visible radiation

2 — Rear face (INNOVATION)	**Metallised Tetra Brik interior (Al + laminated PE)**	**Integrated rear reflector (~85–90% reflectivity). Redirects rearward-escaping radiation back to absorber core. Eliminates the last purchased material from the design.**
3 — Inner shell	50% black matte PET (front face of inner tube)	IR absorption; reduction of front-face reflection
4 — Absorber core	0.1mm aluminium can	Near-instantaneous thermal transfer to working fluid; high conductivity, minimal thermal inertia

Table 3. RTU tube multi-layer architecture. The Tetra Brik rear reflector is a key innovation that completes the 100% recycled design.

INNOVATION — Tetra Brik integrated rear reflector: The interior of Tetra Brik beverage cartons consists of metallised aluminium foil laminated with polyethylene on both faces, providing ~85–90% reflectivity across the visible and near-IR spectrum. When flattened and adhered to the rear face of the inner PET tube — the face without black paint — the aluminium layer faces the absorber core, redirecting radiation that would otherwise escape rearward. This innovation eliminates the only component that in previous versions required purchase (aluminium foil roll), completing the 100% recycled waste design.

4.2 Field validation — experimental data

Test	Condition	Mass	ΔT	Duration	Net power	Efficiency
Field 1	Without insulated reservoir (adverse condition — lower bound)	22 kg	2.7°C	24 min	172.7 W/m²	19.2%
Field 2	Without insulated reservoir (adverse condition — lower bound)	20 kg	7.7°C	59 min	182.1 W/m²	20.2%
Average	—	—	—	—	**177.4 W/m²**	**19.7%**
T max v1	**Static condition (minimum flow rate)**	—	—	—	**66.8°C output**	**lower bound**
T max v2	**Optimised condition April, Córdoba**	—	—	—	**80°C output**	**upper bound**

Table 4. RTU field measurements. Both tests were conducted without an insulated reservoir, meaning the water was losing heat to the environment simultaneously. Measured net power (177 W/m²) is therefore the conservative lower bound of actual collector performance. $Q = m \cdot c \cdot \Delta T$; $P = Q/t$.

METHODOLOGICAL NOTE: Both field measurements were performed without an insulated reservoir — water lost heat to the environment concurrently with solar gain (~40W estimated losses). The net power of 177 W/m² is therefore a **conservative lower bound**. With the refrigerator polyurethane reservoir, losses fall from ~40W to ~5W. All energy impact scenarios in this paper are built on this lower bound and are therefore inherently conservative.

4.3 Comparison with commercial reference systems

System	Max. operating temp.	W/m² thermal	Cost/m²
Commercial flat-plate collector (basic)	60–70°C	400–500W	USD 150–300
RTU v1 — unoptimised	**66.8°C**	**177W (measured)**	**USD 0–5**
RTU v2 — optimised (April)	**80°C**	**>177W (estimated)**	**USD 0–5**
Commercial evacuated tube (premium)	80–120°C	600–700W	USD 400–800

Table 5. Maximum operating temperature comparison. RTU v2 reaches the operational range of premium evacuated tube systems at near-zero material cost.

5. LAYER 2 — THERMAL BATTERY: INSULATED RESERVOIR

The insulated cabinet of a discarded refrigerator serves two simultaneous functions: thermal storage for domestic hot water (Layers 1+2) and evaporator for the ORC cycle (Layer 3). A single waste item resolves both functions. Its injected polyurethane foam ($\lambda \approx$ 0.022–0.028 W/(m·K), 50–80mm thickness) outperforms commercial polyethylene tanks ($\lambda \approx$ 0.035–0.040 W/(m·K)), with overnight losses of only 2–4°C over 12 hours. Volumetric capacity: 180–320 litres per unit. At full load in 200L: 9.3 kWh of stored thermal energy.

WEEE Type	Internal heat exchanger	ORC advantage	Availability
Vertical refrigerator (no-frost)	No-frost aerial serpentine	Very common. Directly reusable.	High
Horizontal chest freezer (Al/Cu plate)	Welded Al/Cu evaporator plate	Direct water contact. Maximum thermal efficiency.	Medium-high
A/C radiator (Cu serpentine)	Copper serpentine with Al fins	λ copper = 385 W/(m·K). Best heat transfer.	Medium

Table 6. WEEE candidates as RTU reservoir/evaporator. Not mutually exclusive variants but potential substitutes depending on locally available waste.

6. ELECTRONIC MANAGEMENT SYSTEM — LAYERS 1+2

Sensor	Location	Measured variable
DS18B20 #1	Collector outlet	Hot water output temperature
DS18B20 #2	Collector inlet	Return temperature to collector
DS18B20 #3	Reservoir bottom	Cold water temperature — ORC critical point
DS18B20 #4	Reservoir top	Internal thermal stratification
PV cell (V_rad)	Collector exterior	Incident solar radiation (V proportional)

Table 7. RTU Phase 1 sensor network.

```
// RTU Firmware v1.0 — Arduino Mega | MIT License
// Fundacion Hijos del Sol y la Basura
// ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
#define DELTA_ON     5.0   // Activate pump: T_col > T_res + 5C
#define DELTA_OFF    2.0   // Stop pump:     T_col < T_res + 2C
#define T_EMERGENCY  90.0  // Force circulation above 90C
#define DELAY_MIN    120   // Anti-flickering delay (seconds)

float T_col = readDS18B20(SENSOR_COLLECTOR);
float T_res = readDS18B20(SENSOR_RESERVOIR);
float duty  = map(analogRead(PV_CELL), 100, 900, 30, 100);

if (!pump && T_col > T_res+DELTA_ON && time>DELAY_MIN)
    startPump(duty); // PWM proportional to solar radiation
if (pump  && T_col < T_res+DELTA_OFF) stopPump();
if (T_col > T_EMERGENCY) { startPump(100); alarm(); }
logSD(ts, T_col, T_res, duty, pump); // dataset for paper
```

7. ENERGY POTENTIAL AND WASTE STREAM ANALYSIS

7.1 Energy per m² of RTU collector — measured data

Parameter	Value	Source
Annual irradiation — Córdoba	2,190 kWh/m²/year	NASA POWER / INTI

Net thermal power (field avg.)	**177 W/m^2 (2 independent tests)**	**Field — Córdoba 2026**
Maximum temperature reached	**80°C (April, unoptimised)**	**Field — Córdoba 2026**
Documented temperature v1	**66.8°C (static condition)**	**Field — Córdoba 2026**
ORC electrical power	**30 W/m^2 continuous (design)**	**Thermodynamic calculation**
Generation per m^2	**720 Wh/day · 262 kWh/year**	**Calculated from ORC design**
Overnight reservoir loss	2–4°C / 12 h	Estimated — PU insulation

Table 8. RTU energy parameters. Green: field-measured. Yellow: calculated from ORC design (pending experimental validation).

7.2 Available waste streams — Argentina (global reference)

Material	Annual production	Possible collector m^2	Bottleneck
PET bottles	3,760 M/year	47.0 M m^2	No
Aluminium cans	**1,410 M/year**	**39.2 M m^2**	**YES — limiting factor**
Tetra Brik cartons	2,440 M/year	30.6 M m^2	No
Discarded refrigerators	600,000/year	600,000 reservoirs	Layer 2 bonus

Table 9. Annual waste stream production in Argentina. Aluminium cans are the limiting factor (scarcest material).

7.3 Impact scenarios — electricity and domestic hot water

Scenario	m^2 installed	Households	kWh/year	Elec. coverage (Argentina)	CO■ avoided
1% potential	392,000	98,000	103 GWh	0.08%	20,000 t
5% potential	1,960,000	490,000	514 GWh	0.4%	103,000 t
20% potential	7,840,000	1,960,000	2.06 TWh	1.6%	412,000 t
100% potential	39,200,000	9,800,000	10.3 TWh	8%	2.06 M t

Table 10. Electrical impact scenarios (30 W/m^2 ORC continuous, 24h). Based on available waste streams in Argentina. CO■ factor: 0.2 kg/kWh.

8. TROPICAL AND GLOBAL POTENTIAL

The RTU system is particularly well-suited to tropical and subtropical regions: higher irradiation, higher ambient temperatures, abundant PET and aluminium waste, and critically low energy access. The system is most powerful exactly where it is most needed.

Region	kWh/m^2/day	W electrical/m^2	Wh/day (10m^2)	vs Córdoba	ORC paybac
Córdoba, Argentina (reference)	**6.0**	**30.0 W**	**7,200**	**×1.00**	**5 months**
NW Argentina / Bolivia	6.8	34.0 W	8,160	×1.13	4 months
NE Brazil	6.5	32.5 W	7,800	×1.08	4 months
Central Mexico	6.4	32.0 W	7,680	×1.07	4 months
Central India	6.9	34.5 W	8,280	×1.15	4 months
Sahel / Sub-Saharan Africa	**7.2**	**36.0 W**	**8,640**	**×1.20**	**3 months**
Central Australia	**7.5**	**37.5 W**	**9,000**	**×1.25**	**3 months**

Table 11. ORC electrical output by climate zone (base: 30W/m^2 at Córdoba, scaled proportionally to irradiation). Payback at USD 0.15/kWh, system cost USD 68.

9. LAYERS 3 AND 4 — ELECTRICITY AND COLD: LOW-TECH ORC WITH ADAPTIVE DYNAMIC EXPANSION

Layers 3 and 4 constitute theoretically grounded design concepts with solid thermodynamic foundations. Their experimental validation requires specific refrigeration engineering expertise and investment in development. Real operational parameters — including minimum functional ΔT and efficiency curves across thermal ranges — are open research variables, not design assumptions. They are presented here as documented working hypotheses.

9.1 The central innovation: adaptive dynamic expansion

A conventional ORC system has a fixed optimal design point: if temperature falls below the threshold, efficiency collapses or the system stops. The RTU innovation resolves this through an adaptive dynamic expansion system governed by solenoid valves and Arduino Nano, which continuously adjusts the active volume of the thermodynamic cycle according to the available thermal differential in real time.

When the reservoir is hot (peak irradiation), the system opens more expansion chambers in parallel, increasing flow and power. As ΔT decreases (night, cloud cover), the Arduino reduces the active volume, maintaining efficient expansion with less fluid. This is what enables the 30 W/m² **continuous** design estimate over 24 hours — not 30W only at peak solar intensity, but a sustained 30W average because the cycle permanently adapts to available conditions.

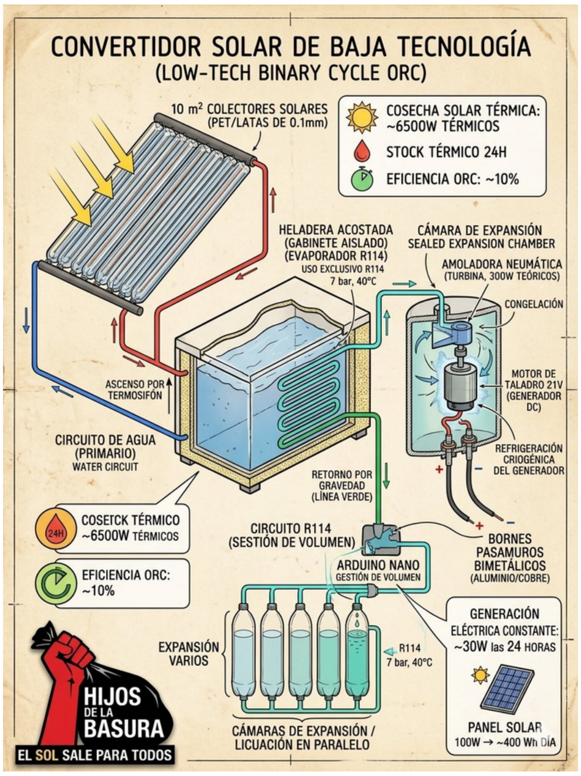

Fig. 4 — ORC diagram v1: R134a cycle, refrigerator as evaporator, pneumatic turbine.

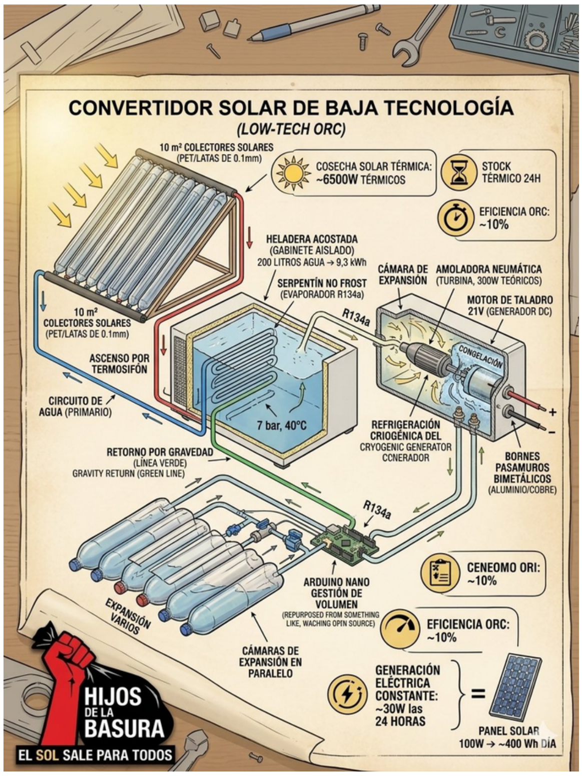

Fig. 5 — ORC diagram v2: submerged no-frost serpentine, parallel expansion chambers.

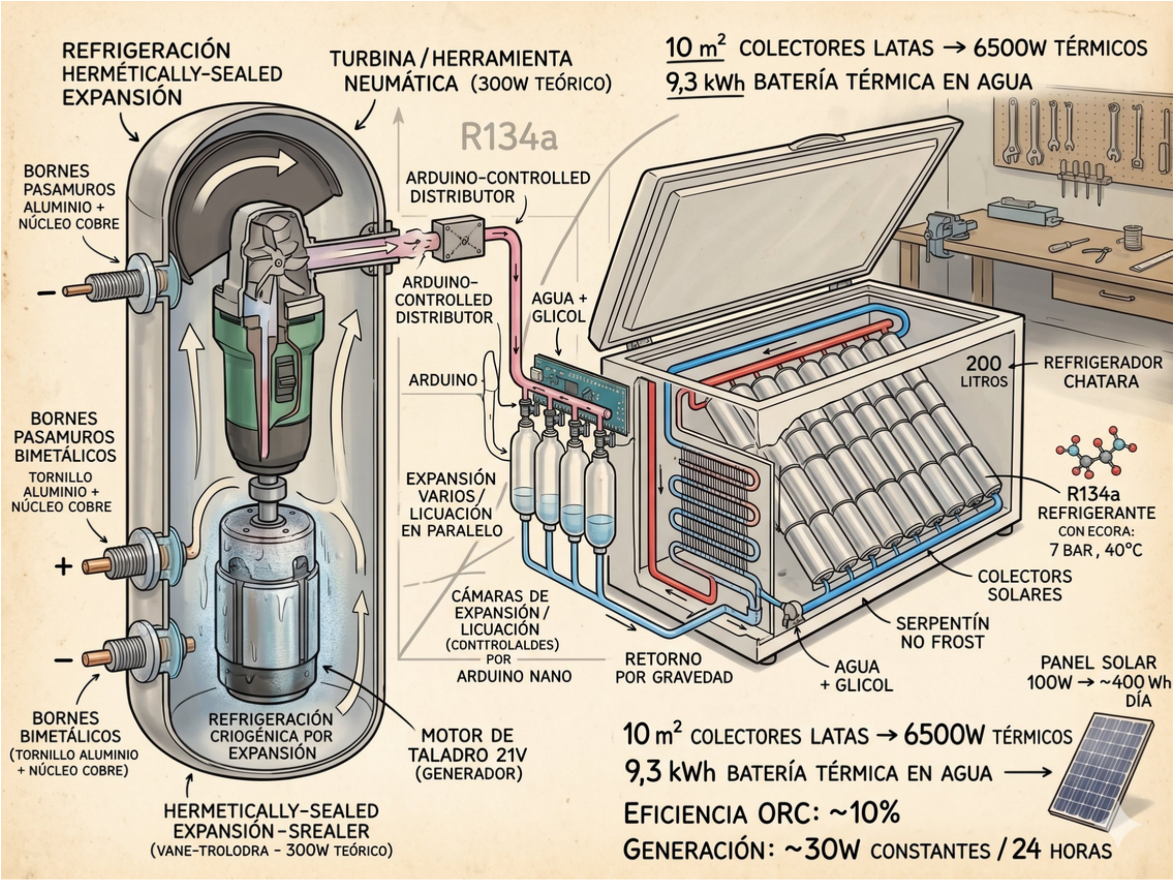

Fig. 6 — ORC detail: hermetically sealed pneumatic grinder/turbine, Arduino Nano as dynamic expansion manager, bimetallic Al/Cu feedthrough terminals, horizontal chest freezer with Al/Cu plate as submerged evaporator.

9.2 ORC cycle components

Component	Material / Origin	Function	Est. cost
Turbine	**Straight pneumatic die grinder (new or salvaged)**	**Fluid expansion → mechanical energy**	**USD 18 new**
DC generator	21V drill motor (WEEE)	Mechanical → electrical (~65% eff.)	USD 0 recycled
Working fluid	R134a refrigerant	Evaporation-expansion cycle 7 bar / 40–80°C	USD 15–25
Evaporator	No-frost serpentine or Al/Cu plate (WEEE)	Heat transfer: water → R134a	USD 0 recycled
Expansion chambers	PET bottles + valves	Volume buffer; dynamic pressure control	USD 5–10
Expansion manager	**Arduino Nano + solenoid valves**	**Continuous active volume adjustment according to available ΔT**	**USD 15–20**
Feedthrough terminals	Al bolt + Cu core bimetallic (fabricated)	Hermetic electrical sealing of generator chamber	USD 5

Table 12. RTU ORC cycle components. Yellow: key innovative components. The pneumatic grinder (USD 18) is viable as a new purchase. All other components are 100% recycled.

9.3 Simultaneous triple energy output

Layer	Output	Direct application	Complexity

1+2 — Heat	**Hot water up to 80°C (validated)**	**Domestic hot water, thermal process applications**	**★■■ One day's work**
3 — Electricity	**30W/m² continuous 24/7 (theoretical design)**	**Lighting, charging, off-grid, data centre infrastructure**	**★★★ Investment required**
4 — Cold	**R134a expansion → residual cold reusable**	**Food refrigeration, passive cooling, server cooling**	**★★★ Integrated into ORC**

Table 13. RTU triple energy output per layer.

10. COST ANALYSIS AND ECONOMIC IMPACT

System	Capacity	USD ref.	24h generation	Payback
Gas water heater	80–150 L	350–700	No	—
Commercial solar basic	150–200 L	1,050–1,750	No	—
Commercial solar premium	200–300 L	2,200–4,400	No	—
100W PV panel + battery	~400 Wh/day	450–700	With battery	18–24 months
RTU Layers 1–2	**80°C · 200L**	**35–70**	**No**	**1–3 months**
RTU Layers 1–2–3	**80°C · 300W/10m²**	**88–115**	**YES 24/7**	**4–5 months**

Table 14. System cost comparison. RTU Layers 1–2–3 is the only system under USD 120 that generates continuous 24/7 electricity without chemical batteries.

The cost per Wh·day of the complete RTU system is up to 7× lower than an equivalent photovoltaic system with battery storage. The advantage is not only economic: it is structural. The RTU requires no lithium or lead-acid batteries — costly, polluting, limited-lifespan components. The RTU battery is the stored hot water itself.

11. LAYER 5 — SOCIAL THERMODYNAMICS

Fig. 7 — Phase 1 complete: Thermal Sovereignty. Phase 2 in design: Electric Sovereignty.

Fig. 8 — RTU prototype in field conditions. 100% urban waste materials.

The four preceding layers demonstrate that the barrier to energy transition in low-income communities is not technological or material: both are resolved by this system. It is organisational.

The fifth layer is not physical. It is the inevitable social consequence of the four preceding layers. Most of humanity does not lack the capacity to work; it lacks access to capital and appropriate technology. The RTU system dramatically reduces the technological barrier required to produce energy infrastructure: materials come from waste, tools are simple and the required training is accessible.

Institutions with territorial presence, training capacity and community service orientation are the natural replication vector for this technology. Three days of training, waste materials and this document are sufficient to deploy energy sovereignty in any sun-accessible community worldwide.

A neighbourhood workshop with access to this documentation can begin manufacturing and selling RTU systems at affordable prices today. It generates local employment, reduces pressure on WEEE management systems and democratises access to energy efficiency. The 600,000 refrigerators discarded annually in Argentina alone are 600,000 thermal batteries waiting to be activated. The billions of cans and PET bottles are collectors waiting to be assembled.

At global scale, tropical regions with high solar irradiation, abundant PET and aluminium waste streams, and low energy access represent the highest-impact market: the RTU system is most powerful exactly where it is most needed.

12. CONCLUSIONS

1. The RTU collector (PET + 0.1mm aluminium can + Tetra Brik) reaches 80°C in April without optimisation — matching premium evacuated tube commercial systems, at near-zero material cost.

2. Measured net thermal power is 177 W/m² under adverse conditions (no insulated reservoir) — a conservative lower bound; actual collector performance is higher.

3. The discarded refrigerator carcass found on public streets provides industrial-grade insulation ($\lambda \approx 0.022$ W/(m·K)) at zero cost.

4. The Tetra Brik rear reflector innovation completes the 100% recycled design, eliminating the last purchased component from the collector.

5. The adaptive dynamic expansion system (Arduino + solenoid valves + modular PET chambers) is the innovation that enables 30 continuous W/m² over 24 hours by permanently adapting the thermodynamic cycle to available ΔT.

6. Layers 3 and 4 (ORC + cold) are theoretically sound designs requiring refrigeration engineering investment for experimental validation.

7. 30 W/m² × 39.2 M possible m² = 1.2 GW continuous = 10.3 TWh/year = 8% of Argentina's total electricity consumption — from currently discarded urban waste.

8. The barrier to popular energy transition is not technological: it is organisational. Layer 5 converts every RTU installation into a local productive node.

TECHNICAL MANIFESTO

Waste is not the absence of value. It is value in disorganised form.

Energy is not merely electricity flowing through a grid: it is heat, storage, motion, cold and organised human work.

Energy sovereignty does not begin at large, inaccessible high-voltage plants. It begins when a community can build, repair and understand its own energy systems.

The Universal Thermodynamic Revolution is born from that conviction.

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